Huu-Truong Le

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Summary

Doctoral researcher in Automatic Control at Université de Lorraine, specializing in AI-driven maintenance scheduling and optimization for complex industrial systems. My research integrates stochastic degradation modeling, graph-based deep reinforcement learning, and optimization methods to address component dependencies and maintenance-induced deterioration. My work combines mathematical modeling, optimization, and machine learning, supported by research experience within the EU-funded MODAPTO project.

Education

- **Université de Lorraine** — Nancy, France
  PhD in Automatic Control, Signal and Image Processing, Computer Engineering — *07/2023 – Expected 11/2026*
  - Funded by: European Union's Horizon Europe programme
  - Thesis title: AI-based machine learning and reasoning for predictive maintenance in Industry 4.0
- **University of Danang - University of Science and Technology** — Danang, Vietnam
  Engineer's degree in Automatic Control — *09/2018 – 09/2022*
  - Ranked 2nd in graduating class, top 3% in major

Research Experience

- **Nancy Research Center for Automatic Control** — Nancy, France
  Doctoral Researcher — *07/2023 – present*
  - Conducted a systematic literature review on maintenance scheduling in Reconfigurable Manufacturing Systems, synthesizing existing approaches and identifying key research gaps in maintenance modeling and decision making.
  - Developed a rolling-horizon, dynamic maintenance grouping framework that incorporates key characteristics of reconfigurable manufacturing systems. Applied a genetic algorithm to optimize maintenance schedules, achieving up to a 5.03% reduction in maintenance costs compared with individual maintenance strategies under dynamic operating conditions.
  - Proposed a state-dependent discrete-state Markov degradation model using directed disassembly graphs and matrices to capture additional deterioration caused by component disassembly operations.
  - Developed a graph-based deep reinforcement learning framework for maintenance scheduling in multi-component systems, integrating GCN and Dueling Double DQN to account for component dependencies and disassembly-induced degradation. Evaluated the approach against four reinforcement learning baselines, achieving up to a 20.3% reduction in long-term maintenance cost rate compared with non-graph methods in simulation.

Publications

- **Manuscripts**
  - **Huu-Truong Le**, Phuc Do & Alexandre Voisin. "Deep graph reinforcement learning–based maintenance scheduling for multi-component systems considering disassembly impact". [planned submission to Reliability Engineering & System Safety in 11/2026]
- **Journals**
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "Maintenance Scheduling in Reconfigurable Manufacturing Systems: A Literature Review and a Dynamic Grouping–Based Framework". *Journal of Risk and Reliability.* [accepted]
- **Conferences**
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "AI-based maintenance scheduling framework considering disassembly impact". 23rd IFAC World Congress, 2026. [accepted]
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "A Framework for Maintenance Scheduling in Reconfigurable Manufacturing Systems". 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability, 2025.
  - Huy Vu Tran, Kim Anh Nguyen, Duc Hanh Dinh, **Huu-Truong Le**, & Phuc Do. "Evaluation of the health state of oil-immersed transformer based on dissolved gas analysis: An empirical study". 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability, 2025.
  - **Huu-Truong Le** , Chiara Franciosi, Phuc Do, & Alexandre Voisin. "Contribution of Maintenance to Reconfigurable Manufacturing Systems: State of the Art and Challenges". 6th IFAC Workshop on Advanced Maintenance Engineering, Services and Technology AMEST, 2024.

Projects

- **MODAPTO** [**website**] 07/2023 – 01/2026
  Repository
  - Contributed to Work Package 5 of the project, focusing on predictive maintenance services and maintenance decision-support for industrial systems.
  - Analyzed industrial data provided by the project partner (SEW USOCOME) to support degradation assessment, maintenance planning, and decision-making requirements.
  - Developed a dynamic grouping maintenance policy to optimize predictive maintenance interventions and minimize total maintenance costs while supporting reliable system operation.

Teaching Experience

- **University of Danang - University of Science and Technology** Danang, Vietnam
  Teaching Assistant *08/2021 – 12/2021*
  - Guided students in project-based learning activities, providing technical support in hardware and software design and implementation.
  - Assisted students with microcontroller programming, PLC programming, and the design of DC motor control circuits.

Professional Experience

- **Ban Vien Corporation** Danang, Vietnam
  Embedded Software Engineer *10/2022 – 02/2023*
  - Developed a SystemC/C++ model of an automotive System-on-Chip (SoC) to support system-level simulation, performance evaluation, and verification of hardware/software interactions during early design stages.
- **Thua Thien Hue Power Company** Hue, Vietnam
  Engineering Intern *04/2022 – 06/2022*
  - Developed practical knowledge of SCADA systems and the IEC 61850 standard through the survey and analysis of the Phu Bai substation, contributing to the evaluation of substation monitoring, control, and communication solutions.

Courses and certifications

- **Courses**: Machine Learning Specialization, Reinforcement Learning Specialization, Graph Neural Network [1] [2]

Skills

- **AI & Machine Learning**: Deep Reinforcement Learning (Value-based, Actor-Critic, GRU, LSTM, ...), Graph Neural Networks (GCN, GAT, Spatio-temporal), Metaheuristics (GA)
- **Maintenance & Reliability**: Condition-based Maintenance, Discrete-state Markov Degradation, Reliability Engineering, Degradation Modeling
- **Programming**: Python (PyTorch, NumPy, Matplotlib, Pandas, ...), C/C++, MATLAB, SQL, Git
- **Languages**: English (C1), French (Beginner), Vietnamese (Native)

References

- **Phuc Do** Email : phuc.do@mines-ales.fr
  Professor - IMT Mines Alès
- **Alexandre Voisin** Email : alexandre.voisin@univ-lorraine.fr
  Professor - Université de Lorraine